

2.4 Unit Tangent, Normal, and Binormal Vectors

Tangent Vectors

Recall that in section 2.2 we discussed the idea of creating tangent lines to smooth vector-valued functions. Specifically, we stated the following definition.

Definition: Suppose that $\mathbf{r}'(t) \neq \mathbf{0}$. The vector $\mathbf{r}'(t)$ is called the tangent vector to the curve C at t .

Therefore, we call, $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$ the unit tangent vector.

Example 1: Find the unit tangent vector to the graph of

$\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ at the point where $t = \frac{\pi}{4}$.

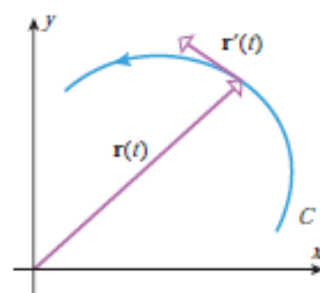
We calculate that $\mathbf{r}'(t) = -\sin(t)\mathbf{i} + \cos(t)\mathbf{j} + \mathbf{k}$.

Therefore, $\mathbf{r}'\left(\frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right)\mathbf{i} + \cos\left(\frac{\pi}{4}\right)\mathbf{j} + \mathbf{k} = -\frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}$ represents the

tangent vector to $\mathbf{r}(t)$.

So, $\mathbf{T}\left(\frac{\pi}{4}\right) = \frac{\mathbf{r}'\left(\frac{\pi}{4}\right)}{\|\mathbf{r}'\left(\frac{\pi}{4}\right)\|} = \frac{-\frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}}{\sqrt{\left(-\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2 + (1)^2}} = \frac{-\frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}}{\sqrt{2}} = -\frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$ is the unit tangent

vector.



Normal Vectors

Recall that in section 2.2 we also introduced the following theorem.

Theorem: If $\mathbf{r}(t)$ is a differentiable vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t , then

$$\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$$

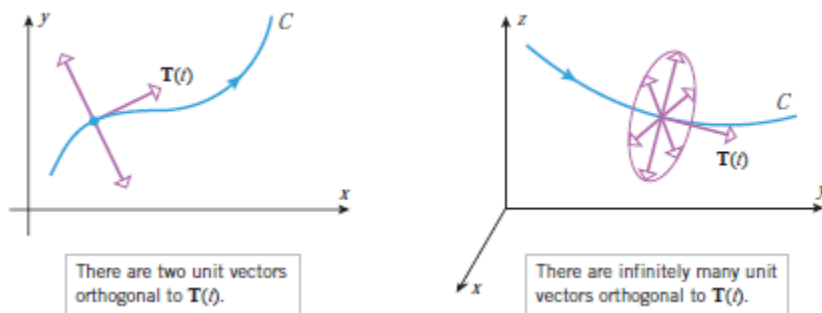
that is, $\mathbf{r}(t)$ and $\mathbf{r}'(t)$ are orthogonal vectors for all t .

Using this theorem we can conclude that since $\mathbf{T}(t)$ has a constant norm (that is, $\|\mathbf{T}(t)\|$ is constant) then $\mathbf{T}'(t)$ is orthogonal to it. This leads us to the following definitions.

Definition: Provided that $\mathbf{T}'(t) \neq \mathbf{0}$, we call $\mathbf{T}'(t)$ a normal vector to the curve C at t .

Therefore, we call $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$ the principle unit normal vector to C at t .

Notice that there exist multiple normal vectors to a curve C and a given point t . Therefore, the idea of the principle unit normal vector is specifying a specific vector out of these several possibilities.



Note: As with the principle square root, if we are asked to find *the* unit normal vector, it is assumed to be the principle unit normal vector.

Example 2: Find the unit normal vector to $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ at $t = \frac{\pi}{4}$.

This is the same curve from Example 1. So we already know that $\mathbf{T}\left(\frac{\pi}{4}\right) = -\frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$, but we need to find $\mathbf{T}(t)$ so that we can also find $\mathbf{T}'(t)$ and therefore $\mathbf{T}'\left(\frac{\pi}{4}\right)$.

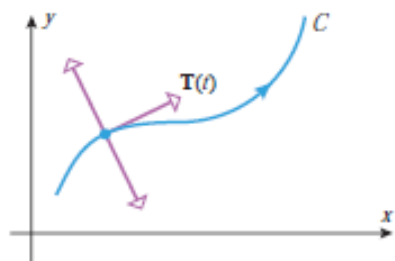
We know that $\mathbf{r}'(t) = -\sin(t)\mathbf{i} + \cos(t)\mathbf{j} + \mathbf{k}$ and thus $\|\mathbf{r}'(t)\| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$.

Therefore, $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \frac{-\sin(t)\mathbf{i} + \cos(t)\mathbf{j} + \mathbf{k}}{\sqrt{2}} = -\frac{1}{\sqrt{2}}\sin(t)\mathbf{i} + \frac{1}{\sqrt{2}}\cos(t)\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$.

So, $\mathbf{T}'(t) = -\frac{1}{\sqrt{2}}\cos(t)\mathbf{i} - \frac{1}{\sqrt{2}}\sin(t)\mathbf{j} + 0\mathbf{k}$ which gives

$$\mathbf{T}'\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}\cos\left(\frac{\pi}{4}\right)\mathbf{i} - \frac{1}{\sqrt{2}}\sin\left(\frac{\pi}{4}\right)\mathbf{j} + 0\mathbf{k} = -\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + 0\mathbf{k}$$

$$\text{Thus, } \mathbf{N}\left(\frac{\pi}{4}\right) = \frac{\mathbf{T}'\left(\frac{\pi}{4}\right)}{\|\mathbf{T}'\left(\frac{\pi}{4}\right)\|} = \frac{-\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + 0\mathbf{k}}{\sqrt{\frac{1}{4} + \frac{1}{4}}} = \frac{-\frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + 0\mathbf{k}}{\frac{1}{\sqrt{2}}} = -\frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + 0\mathbf{k}$$



As we mentioned above there are actually many different normal vectors to a given curve. As shown above in 2-space there exists exactly 2 normal vectors to the curve C at the point t .

Theorem: Let C be a smooth curve in 2-space. The principle unit normal vector at each point along C is the unit normal vector that points *inward* (towards the concave side of the curve).

Proof (not done in class):

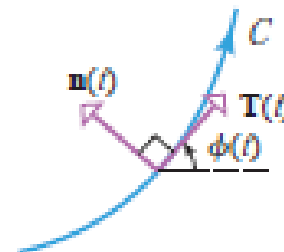
Let $\phi(t)$ be the angle between the positive x -axis and $\mathbf{T}(t)$, $0 \leq \phi(t) \leq 2\pi$.

Let $\mathbf{n}(t)$ be the vector obtained by rotating $\mathbf{T}(t)$ counter-clockwise by $\frac{\pi}{2}$ radians.

Since $\mathbf{T}(t)$ is a unit vector we can express it as $\mathbf{T}(t) = \cos(\phi(t))\mathbf{i} + \sin(\phi(t))\mathbf{j}$

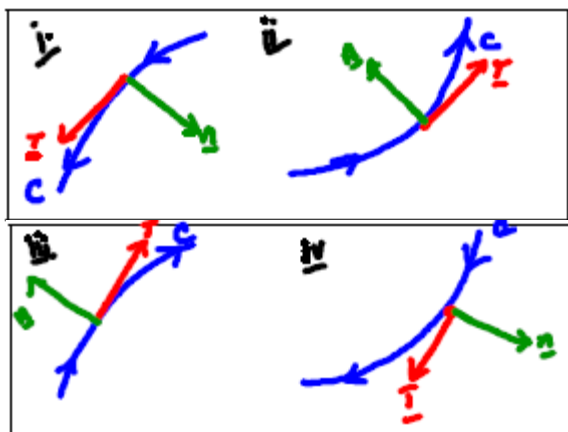
and since $\mathbf{n}(t)$ is just a rotation of $\mathbf{T}(t)$ we can express $\mathbf{n}(t)$ as:

$$\begin{aligned}\mathbf{n}(t) &= \cos\left(\phi(t) + \frac{\pi}{2}\right)\mathbf{i} + \sin\left(\phi(t) + \frac{\pi}{2}\right)\mathbf{j} \\ &= \left[\cos(\phi(t))\cos\left(\frac{\pi}{2}\right) - \sin(\phi(t))\sin\left(\frac{\pi}{2}\right)\right]\mathbf{i} + \left[\sin(\phi(t))\cos\left(\frac{\pi}{2}\right) + \cos(\phi(t))\sin\left(\frac{\pi}{2}\right)\right]\mathbf{j} \\ &= -\sin(\phi(t))\mathbf{i} + \cos(\phi(t))\mathbf{j}\end{aligned}$$



Clearly $\mathbf{n}(t)$ represents a normal vector to the curve C since it was a $\frac{\pi}{2}$ radians rotation of a tangent vector, but we also know that $\mathbf{T}'(t)$ is a normal vector to C . Interestingly we see that:

$$\mathbf{T}'(t) = -\sin(\phi(t))\frac{d\phi}{dt}\mathbf{i} + \cos(\phi(t))\frac{d\phi}{dt}\mathbf{j} = [-\sin(\phi(t))\mathbf{i} + \cos(\phi(t))\mathbf{j}]\frac{d\phi}{dt} = \mathbf{n}(t)\frac{d\phi}{dt}$$



So, if $\frac{d\phi}{dt} > 0$ (i.e., ϕ is increasing) then $\mathbf{n}(t)$ and $\mathbf{T}'(t)$ point in the same direction.

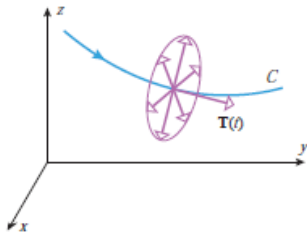
If $\frac{d\phi}{dt} < 0$ (i.e., ϕ is decreasing) then $\mathbf{n}(t)$ and $\mathbf{T}'(t)$ point in opposite directions.

Therefore, since $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$ it always points towards the concave side of the curve.

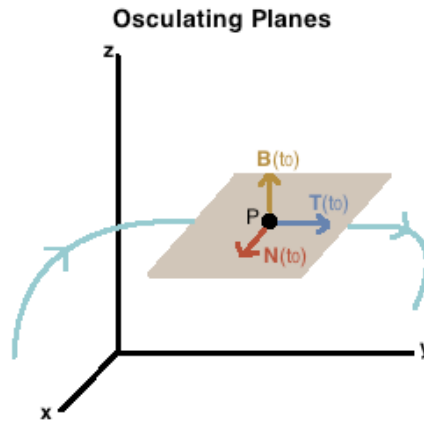
Note: In 2-space $\mathbf{N}(t)$ is commonly referred to as the inward unit normal.



Binormal Vectors



In 3-space there are infinitely many normal vectors at a point on C . $\mathbf{N}(t)$ is just one of these vectors, specifically the vector that when paired with $\mathbf{T}(t)$ determine a plane which is tangent to C . This plane is called the osculating plane.



Definition: If C is a graph of a vector-valued function $\mathbf{r}(t)$ in 3-space, then we define the binormal vector to C at t to be

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Note: $\mathbf{B}(t)$ is a unit vector.

Example 3: Prove that $\mathbf{B}(t)$ is a unit vector.

We see that $\|\mathbf{B}(t)\| = \|\mathbf{T}(t) \times \mathbf{N}(t)\|$ and we know that $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin(\theta)$ where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$. Since $\mathbf{T}(t)$ and $\mathbf{N}(t)$ are perpendicular we know that $\theta = \frac{\pi}{2}$ and therefore we have:

$$\|\mathbf{B}(t)\| = \|\mathbf{T}(t) \times \mathbf{N}(t)\| = \|\mathbf{T}(t)\| \|\mathbf{N}(t)\| \sin\left(\frac{\pi}{2}\right) = (1)(1)(1) = 1$$

Therefore, $\mathbf{B}(t)$ is a unit vector.

Definition: A coordinate system determined by $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ is called the TNB-frame or the Frenet-frame.

Theorem: The binormal vector $\mathbf{B}(t)$ can be expressed directly in terms of $\mathbf{r}(t)$ as:

$$\mathbf{B}(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}$$

Proof: Follows from the definitions of $\mathbf{B}(t)$, $\mathbf{N}(t)$, and $\mathbf{T}(t)$.

Example 4: Find the binormal vector to $\mathbf{r}(t) = \cos(t)\mathbf{i} + \sin(t)\mathbf{j} + t\mathbf{k}$ at $t = \frac{\pi}{4}$.

We plan to use the above formula to calculate $\mathbf{B}\left(\frac{\pi}{4}\right)$,

First we compute the following:

$$\mathbf{r}'\left(\frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right)\mathbf{i} + \cos\left(\frac{\pi}{4}\right)\mathbf{j} + \mathbf{k} = -\frac{\sqrt{2}}{2}\mathbf{i} + \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}$$

$$\mathbf{r}''\left(\frac{\pi}{4}\right) = -\cos\left(\frac{\pi}{4}\right)\mathbf{i} - \sin\left(\frac{\pi}{4}\right)\mathbf{j} + 0\mathbf{k} = -\frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + 0\mathbf{k}$$

$$\mathbf{r}'\left(\frac{\pi}{4}\right) \times \mathbf{r}''\left(\frac{\pi}{4}\right) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 1 \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{vmatrix} = \frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}$$

$$\left\| \mathbf{r}'\left(\frac{\pi}{4}\right) \times \mathbf{r}''\left(\frac{\pi}{4}\right) \right\| = \sqrt{\frac{1}{2} + \frac{1}{2} + 1} = \sqrt{2}$$

$$\text{Therefore, } \mathbf{B}\left(\frac{\pi}{4}\right) = \frac{\mathbf{r}'\left(\frac{\pi}{4}\right) \times \mathbf{r}''\left(\frac{\pi}{4}\right)}{\left\| \mathbf{r}'\left(\frac{\pi}{4}\right) \times \mathbf{r}''\left(\frac{\pi}{4}\right) \right\|} = \frac{\frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}}{\sqrt{2}} = \frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$$

OR

From example 1 and 2 we know that $\mathbf{T}\left(\frac{\pi}{4}\right) = -\frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} + \frac{1}{\sqrt{2}}\mathbf{k}$ and $\mathbf{N}\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + 0\mathbf{k}$.

$$\text{Therefore } \mathbf{B}\left(\frac{\pi}{4}\right) = \mathbf{T}\left(\frac{\pi}{4}\right) \times \mathbf{N}\left(\frac{\pi}{4}\right) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \end{vmatrix} = \frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + \left(\frac{\sqrt{2}}{4} + \frac{\sqrt{2}}{4}\right)\mathbf{k} = \frac{1}{2}\mathbf{i} - \frac{1}{2}\mathbf{j} + \frac{\sqrt{2}}{2}\mathbf{k}$$

Note: Both approaches to calculating $\mathbf{B}(t)$ produce the same result.